

Task 4 Report – Advanced Analytics (Basic)

Internship: ApexPlanet Software Pvt. Ltd. – Data Analytics Internship

Objective: Perform statistical analysis, customer segmentation, and build a basic predictive model using Python and Jupyter Notebook.

Tools Used: Jupyter Notebook, Python, Pandas, Matplotlib, SciPy, Scikit-learn.

Dataset: Sample Superstore Dataset.

Work Performed:

1. Statistical Analysis (Mean, Median, Mode, Standard Deviation, Skewness).
2. Hypothesis Testing using an independent t-test.
3. Customer Segmentation using StandardScaler, K-Means, and PCA visualization.
4. Predictive Modeling using Linear Regression with train/test split and evaluation using R^2 , MAE, and RMSE.

Key Outcomes:

- Completed descriptive statistical analysis.
- Performed hypothesis testing.
- Created four customer clusters using K-Means.
- Visualized clusters using PCA.
- Built and evaluated a Linear Regression model.

Conclusion:

Task 4 provided practical experience in statistics, machine learning, clustering, and predictive analytics using Python.